Cascade Failure & Load Shedding in Powergrids

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Motivation/Research Question

- Cascades occur when flows reroute → overloads → additional trips → fragmentation / load loss
- Cascading failures can propagate far beyond the initial outage
- Operators mitigate by load shedding

Research Questions:

- Which nodes/edges are most critical under cascade dynamics?
- Do centrality metrics predict true criticality?
- Do the conclusions hold across case118 and case300?

Dataset / system overview

Case118: 118 buses
173 lines
99 loads
53 generators
4.48% baseline max loading

- baseline test system
- used for method development and analysis

Case300: 300 buses
281 lines
193 loads
68 generators
8.73% baseline max loading

- larger validation system
- used to test whether patterns scale

Cascade Simulation Pipeline

1. **Initialize**
   Remove one line from the IEEE 118-bus or 300-bus network
2. **Power flow**
   Run AC power flow (pandapower) to compute new line loadings
3. **Overload check**
   Flag lines exceeding threshold: baseline × (1 + α), α = 0.2
4. **Trip & repeat**
   Disable overloaded lines, loop until no new trips or solver fails
5. **Measure impact**
   Record cascade size, load shed %, and LCC size per contingency

Simulation parameters

Networks	IEEE 118-bus & 300-bus
Lines tested (118)	All 173 lines (N-1)
Lines tested (300)	All 411 lines (N-1)
Alpha (α)	0.2 (20% overload margin)
Min base loading	0.1% (floor for near-zero baselines)
Solver	pandapower AC power flow
Graph metrics	Degree, betweenness, closeness
Output metrics	Cascade size, shed %, LCC size
Reproducibility	Fixed seed, saved configs, JSON output

Overload threshold rule
load > baseline × (1 + α), with min floor of 0.1%

Graph-Based Criticality Metrics

For nodes:

- Degree
- Betweenness
- Closeness
- Eigenvector centrality

For edges:

- Edge betweenness

These metrics were compared against simulation outcomes such as load shed and cascade size.

Metric	Intuition
Degree	Local connectivity
Betweenness	Bridge-like importance
Closeness	Global accessibility
Eigenvector	Influence through important neighbors
Edge betweenness	Flow bottleneck criticality

Implementation Progress

Task	Status	Details
Data ingestion + baseline model setup	Done	case118 + case300 via pandapower
Graph construction + centrality computation	Done	Degree, betweenness, closeness, eigenvector centrality, edge betweenness from graph representations
Contingency and cascade simulator	Done	Iterative AC power flow, overload tripping
Preliminary results on case118	Done	All 173 N-1 contingencies simulated
Comparative analysis (graph vs simulation)	Done	Compared graph metrics against cascade outcomes using load shed %, cascade size, correlations, and rank overlap
Threshold calibration / sensitivity analysis	Done	Performed parameter sweeps over alpha and min_base_loading to test cascade sensitivity
Case300 validation	Done	Extended the full pipeline to the IEEE 300-bus network and ran all 411 single-line contingencies
Final visualizations + presentation refinement	Done	Produced summary tables, scatter plots, heatmaps, and ranked criticality results for case118 and case300

Case118: Graph Structure Results

Top 10 buses by degree:

Bus	Degree
48	9
99	8
11	7
76	6
79	6
91	6
14	5
16	5
31	5
36	5

Top 10 buses by betweenness:

Bus	Betweenness
76	0.356
48	0.348
68	0.346
79	0.220
69	0.210
23	0.185
22	0.182
99	0.174
41	0.155
39	0.144

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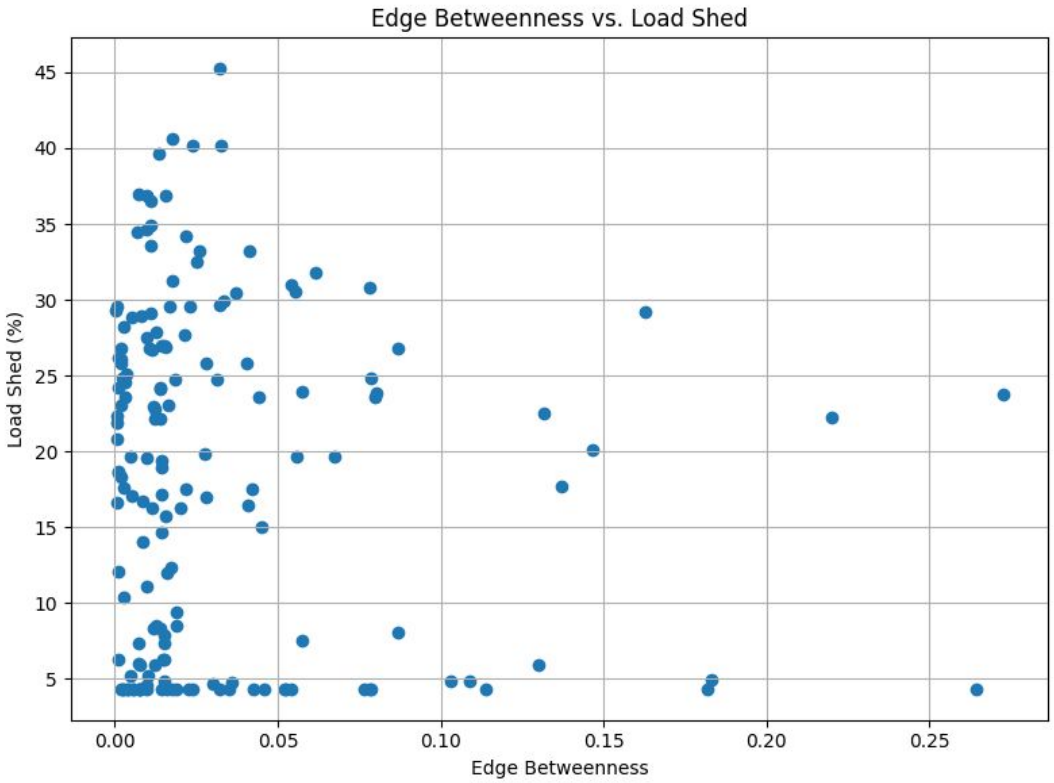
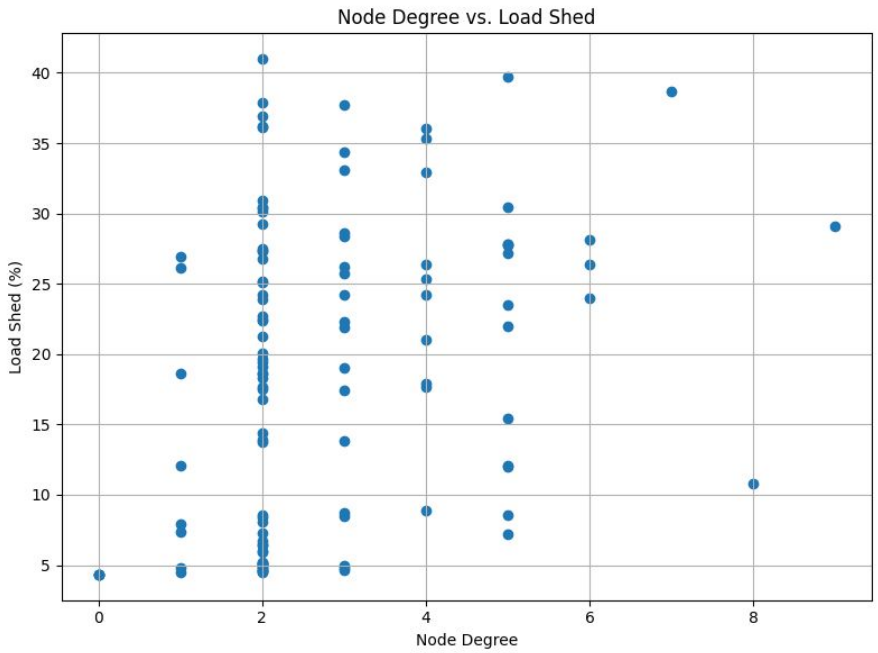
Case118: Worst Cascades

Initial Line	Cascade Size	Shed %	LCC
31	109	22.25	11
0	99	21.10	11
39	102	19.31	13
19	99	18.67	23
141	87	18.10	15
115	89	18.06	29

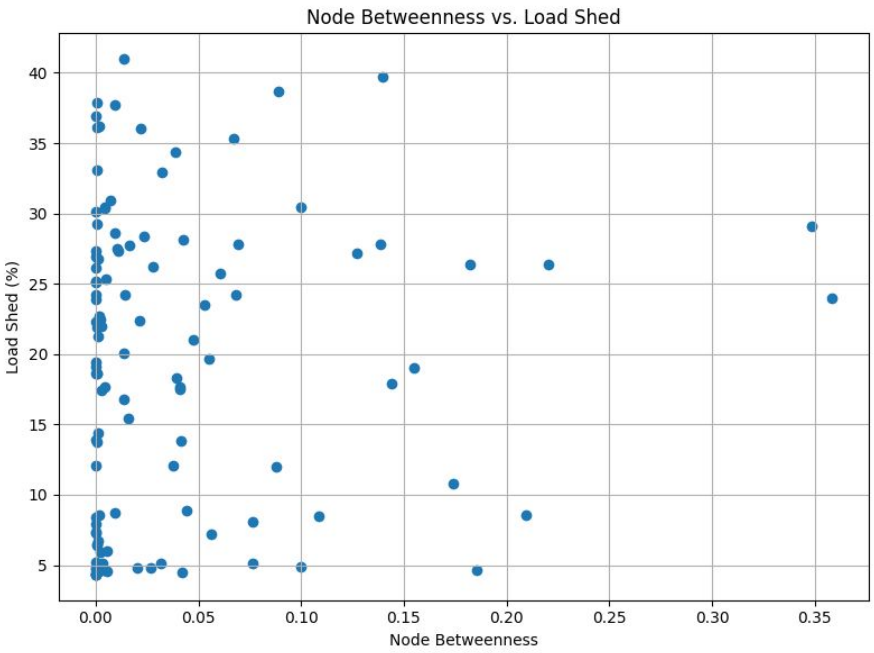
Case118: Graph vs Simulation



Case118: Node Metrics vs Simulation Impact



Pearson = 0.273
Spearman = 0.306

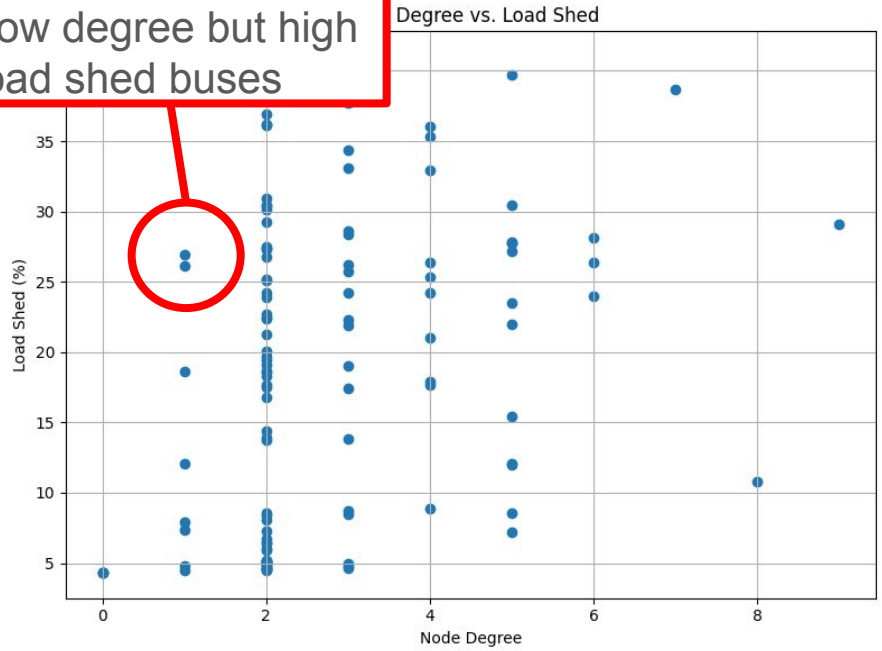


Pearson = 0.072
Spearman = 0.179

Node degree appears to be a stronger predictor of simulation impact than node betweenness, but neither metric fully explains cascade severity.

Case118: Node Metrics vs Simulation Impact

Low degree but high load shed buses



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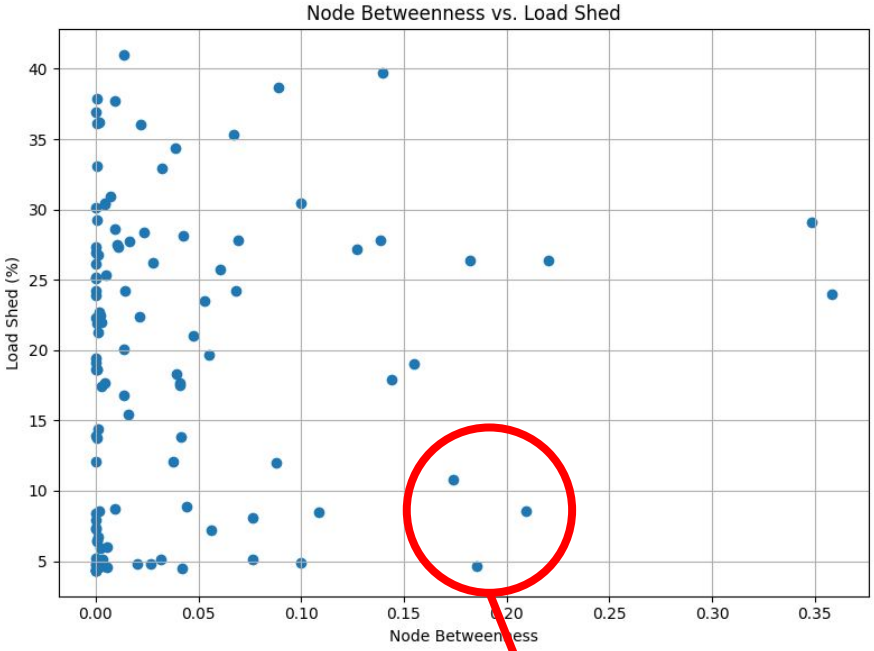


Pearson = 0.072

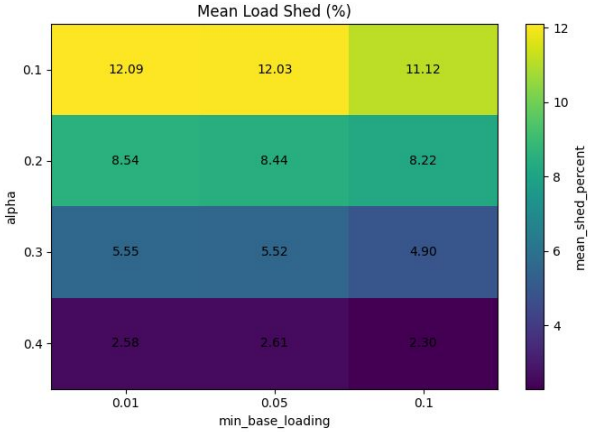
high-betweenness but only moderate load shed buses

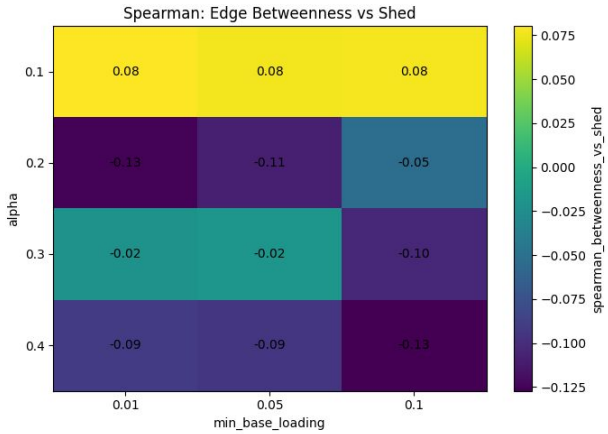
Node degree appears to be a stronger predictor of simulation impact t her metric fully explains cascade severity.

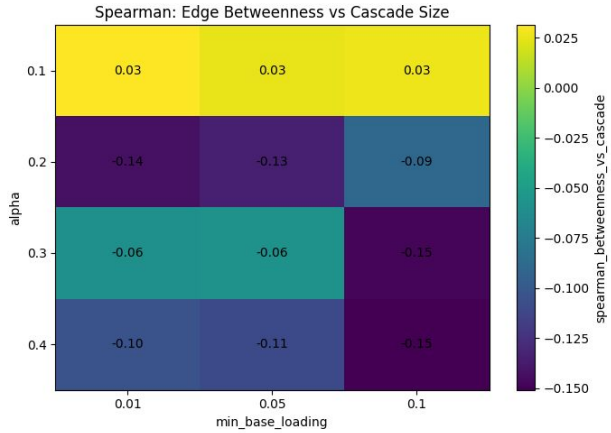
Case118: Node Metrics vs Simulation Impact

Dataset	Metric	Pearson vs Shed	Spearman vs Shed	Pearson vs Cascade Size	Spearman vs Cascade Size	Top-10 Overlap vs Shed	Top-10 Overlap vs Cascade
case118	node betweenness	0.07199	0.17852	0.13299	0.28679	3	2
case118	degree	0.27321	0.30618	0.40564	0.55613	3	3
case300	node betweenness	0.124	0.62602	.545	0.53109	8	7
case300	degree	0.406	0.74716	.282	0.67684	8	7

Case118: Threshold Calibration Results







Takeaways:

- Lower α leads to more severe cascades
- As α increases, cascades become less severe.
- Edge betweenness remains a weak predictor of load shed and cascade size
- Changing min_base_loading has only a modest effect compared with changing α.

Case300: Validation Setup

- Same pipeline applied to larger network
- Used to test whether observed trends from case118 generalize
- More buses, more lines, more complex failure interactions

Case300: Graph Structure Results

Top 10 buses by degree:

Bus	Degree
118	7
63	7
30	7
108	6
115	6
189	6
111	5
78	5
34	5
188	4

Top 10 buses by betweenness:

Bus	Betweenness
40	0.0351
60	0.0334
37	0.0263
30	0.0250
14	0.0245
13	0.0232
7	0.0225
36	0.0216
10	0.0186
65	0.0185

Case300: Graph Structure Results

Top 10 buses by degree:

Bus	Degree
118	7
63	7
30	7
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111	5
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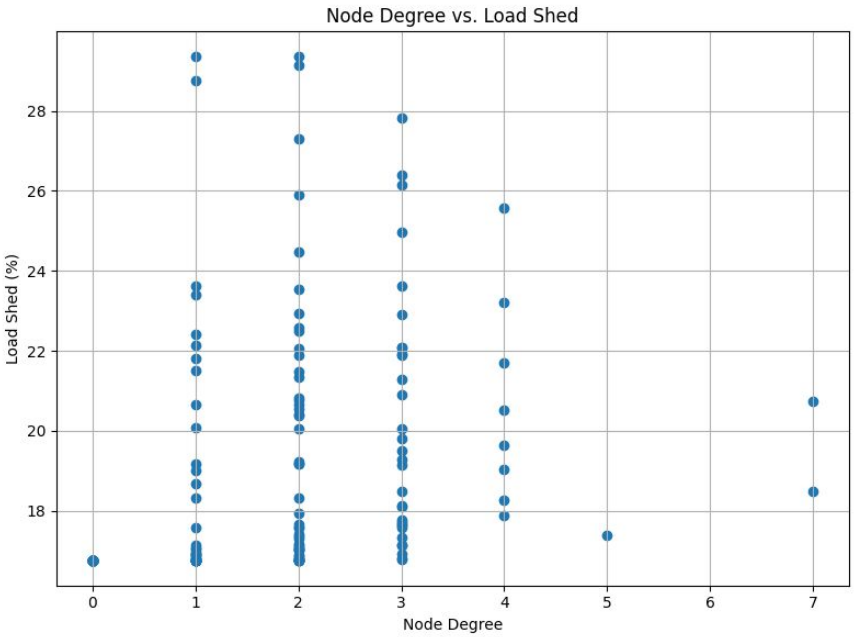
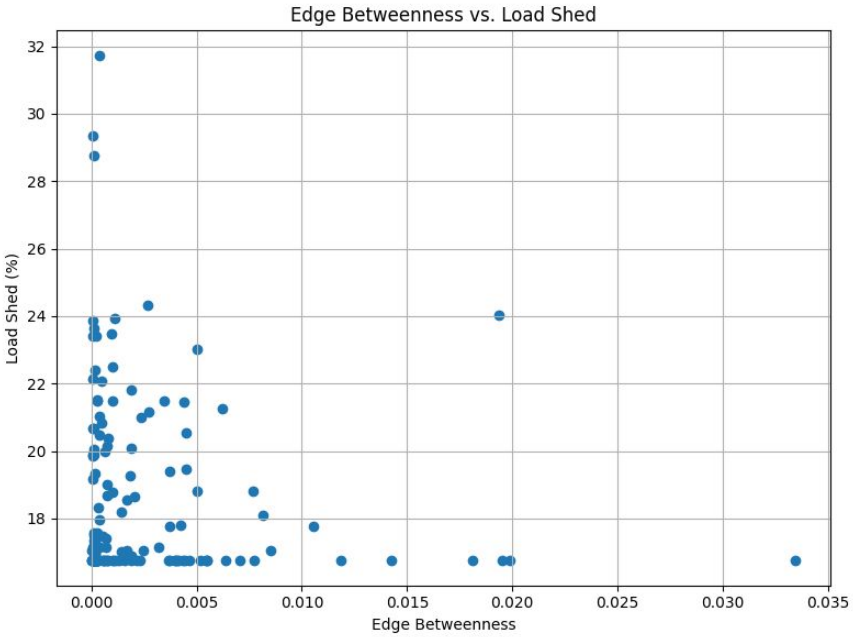
13	0.0232
7	0.0225
36	0.0216
10	0.0186
65	0.0185

Case300: Worst Cascades

Triggered Line	Cascade Size	LCC Size	Shed %
225	79	30	31.72
201	64	28	29.35
147	66	28	28.77
228	69	34	26.27
269	74	34	26.26
263	42	68	24.49
177	74	25	24.33
99	81	34	24.02

Case300: Graph vs. Simulation





Case300: Node Metric Correlations

Metric	Pearson vs Load Shed	Spearman vs Load Shed	Pearson vs Cascade Size	Spearman vs Cascade Size
Degree	**0.406**	**0.747**	**0.545**	**0.794**
Closeness	0.227	0.676	0.395	0.717
Eigenvector	-0.047	0.659	0.048	0.701
Node Betweenness	0.124	0.626	0.282	0.648

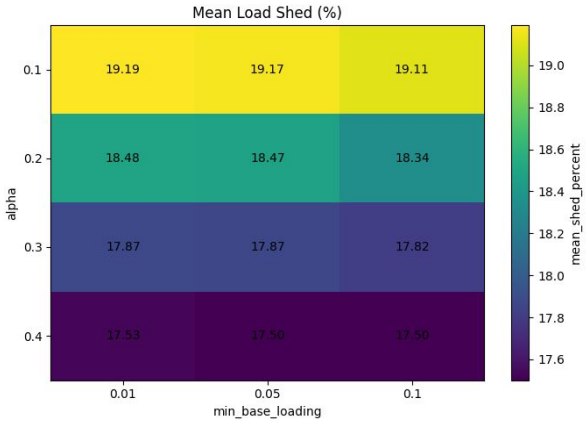
Best Predictor
Strongest metric: Degree

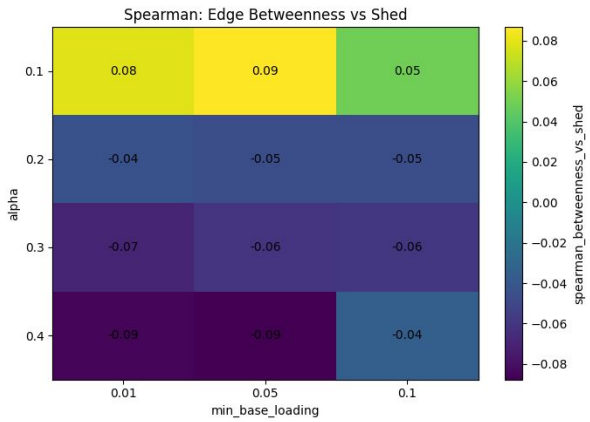
Spearman vs load shed: 0.747
Spearman vs cascade size: 0.794

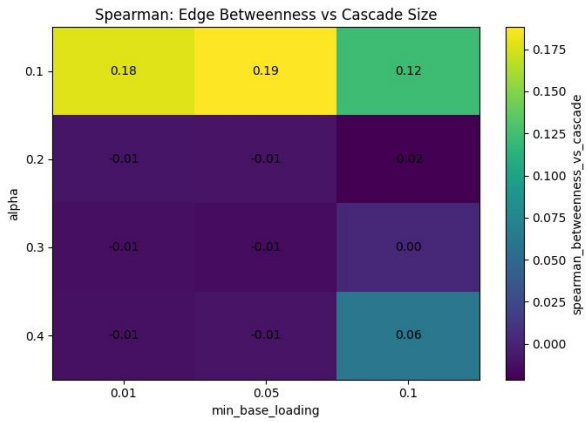
Broader pattern
Closeness, eigenvector, and node betweenness correlate strongly with impact.

Case300 shows a clearer link between graph structure and cascade severity.

Case300: Threshold Calibration Results







- Increasing alpha makes the model less aggressive
- Results are not very sensitive to min_base_loading
- Edge betweenness remains a weak predictor of shed and cascade size across most settings
- This suggests threshold calibration changes severity more than it improves graph-vs-simulation agreement

Case118 vs Case300: Cross-Network Comparison

CASE118 — NETWORK STATS

Buses / Lines 118 / 173	Graph edges 166
Max load shed 45.2%	Max cascade size 109 lines

Top critical node (betweenness)
Bus 76 — betweenness 0.358, degree 6

Top critical edge (betweenness)
Line 109, Bus 68→76 — betweenness 0.273

CASE300 — NETWORK STATS

Buses / Lines 300 / 283	Graph edges 281
Max load shed 31.7%	Max cascade size 85 lines

Top critical node (betweenness)
Bus 40 — betweenness 0.035, degree 3

Top critical edge (betweenness)
Line 73, Bus 40→60 — betweenness 0.033

Case118 vs Case300: Cross-Network Comparison

GRAPH METRIC CORRELATIONS WITH LOAD SHED % (SPEARMAN)

Metric	118 vs shed	300 vs shed	118 vs cascade	300 vs cascade
Degree	+0.367	+0.747	+0.311	+0.794
Node betweenness	+0.184	+0.626	+0.172	+0.648
Closeness	−0.084	+0.676	−0.138	+0.717
Eigenvector	−0.043	+0.659	−0.086	+0.701

CASE118 KEY INSIGHT

Degree is the best predictor (r = 0.37) but correlations are generally weak — graph metrics alone cannot identify the most dangerous lines.

CASE300 KEY INSIGHT

Correlations are substantially stronger across all metrics (r = 0.63–0.79) — graph structure is a better predictor of cascade impact in larger, denser networks.

Final Conclusions and Takeaways

Criticality is not captured equally by all graph metrics.
Degree was the strongest node-level overall, while edge betweenness was weak for predicting cascade severity in case118.

Simulation-based criticality and graph-based importance do not fully match.
In case118, several low-degree buses still caused large load shed, and some high-betweenness components were only moderately damaging.

The relationship strengthens in case300.
In the larger network, node metrics correlate much more strongly with load shed and cascade size, suggesting topology is more predictive there.

How well do these results align with prior literature on cascade vulnerability, and do the same patterns hold under more realistic outage models?